

Enhanced English Image Captioning Using Attention-Based Deep Learning for Accurate Visual-Linguistic Understanding

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ABSTRACT

The task of English image captioning is particularly difficult because of visual scene complexity, natural language generation nuances and the requirement of accurate visual-semantic grounding. To address them, we present an attention-based deep learning model that employs effective convolutional neural network (CNN) encoders and sequential decoders to generate high-quality English captions. Visual features of state-of-the-art CNN backbones, such as VGG16, InceptionV3, ResNet50, and EfficientNet (B0 -B7) are integrated in our architecture. The multi-scale features extracted are united via attention mechanisms that dynamically match visual representations with the generated language tokens. This improves the capacity of the model in attaining contextual information and semantic nuances in various domains of images. The language modelling section combines attention-based LSTM and GRU networks to generate fluent, grammatically correct, and context-aware sentences. Attention-weighted feature selection reduces lexical hallucinations while strengthening factual grounding in the system. Our goal is a BLEU-4 of over 19% and 84% points gain in semantic accuracy compared to baseline encoder-decoder models. Experiments on large-scale benchmark datasets show that our model beats previous methods, provides better visual-linguistic integration, and is reliable in real applications. These findings represent a significant step forward in the development of smarter and more human-like image captioning in English.

1. Introduction

Image captioning is a branch of computer vision and natural language processing (NLP) that trains robots to write descriptions of visual media in a human-friendly manner. Recent advances in deep learning have improved caption quality, but there are still issues with fine-grained visual-linguistic alignment, contextual relevance, and grammatical accuracy, particularly in complex scenarios. Encoder-decoder architectures have been demonstrated to struggle with modelling delicate spatial interactions and producing captions in context, resulting in lexical hallucinations (irrelevant word predictions). To mitigate these shortcomings, we propose an attention deep learning model, which combines:

- Multi-CNN Feature Extraction: Hybrid encoders (VGG16, InceptionV3, ResNet50, EfficientNet B0-B7) to encode visual features hierarchically at different scales.
- Hierarchical Attention Mechanisms: Dynamically boost regions of the image when generating captions, requiring syntactic and semantic consistency.
- Sequential Language Decoding: LSTM/GRU networks that are conditional on attention-weighted features to generate grammatically structured captions.

We have contributed towards:

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- An innovative combination of CNN backbones to strong visual representation.
- Adaptive attention layers to narrow semantic gaps between images and textual information
- Quantitative gains beyond Transformer-based baselines (e.g., greater BLEU, METEOR, and CIDER scores).

Flickr30k Flickr8K examines confirm our model's advantages in terms of accuracy and efficiency, demonstrating its potential for usage in real-world settings such as automated content tagging and aid for the visually handicapped.

2. Literature Review

Image caption creation is a tough task that requires creating written descriptions for photographs. This task has gained attention in multiple disciplines, including picture retrieval, description, and multimedia applications. This project is challenging due to the complicated nature of visual material and natural language semantics. Deep learning approaches, such as Convolutional Neural Networks (CNNs) for image feature extraction and Long Short-Term Memory (LSTM) networks for caption feature extraction, have become popular solutions to this complex challenge [1]. The encoder-decoder structure, modelled after machine translation, is also utilised to generate descriptions for image captions [2]. To extract visual information, CNN is employed as the encoder and LSTM as the decoder to generate description sentences. The LSTM network is a

distinct RNN structure that is widely employed to generate caption words due to its capacity to convey the gradient descent effect with long-term memory [3]. Gradient vanishing or bursting remains a concern; despite numerous strategies devised to address this, we have yet to identify the ideal configuration for LSTM. In computer vision and image retrieval, it's crucial to use appropriate language to describe images. This is known as automatic annotation of photographs. Using an encoder-decoder architecture is an effective technique to generate captions for photographs. This approach leverages pre-trained CNN models (e.g., Alexnet, VGG16, DenseNet) to extract image attributes, followed by caption creation using recurrent neural networks [4, 5]. Although encoder-decoder models have achieved success, they faced challenges when dealing with complicated scenes that include several objects or complex relationships. This challenge prompted the incorporation of attention mechanisms, which are modeled after human visual perception. By using attention layers, the model can selectively concentrate on particular areas of an image during the generation of each word, enhancing both the relevance and contextual precision of the captions [1]. Expanding on these principles, various models (such as BLEU, ROUGE-L, METEOR) were examined by using a Vision Transformer model to substitute CNN for visual feature extraction. This was combined with a memory-augmented transformer for producing captions that utilize both low-level and high-level image features while integrating prior knowledge through memory vectors [6].

3. Related Works

Melg, Slagh, and Hoxha et al. (2020) [7] According to the authors of "A New CNN-RNN Framework For Remote Sensing Image Captioning," creating thorough and comprehensible descriptions for remote sensing images is a difficult task. They suggested a hybrid strategy that combines captioning for images based on retrieval and production. Their method generates many captions using a bespoke CNN-RNN model with beam search, choosing the best one based on lexical similarity with reference captions for similar photos. This addresses the shortcomings of both approaches when used in separate settings and enhances the quality of such captions. Iyer, Sharmila, and Sudhakar (2022) [8] The use of deep system-neural networks for image captioning is explained in detail in the article "Image Caption Generation using Deep Neural Networks". The authors' approach used CNN-extracted features to create captions using RNNs, which were subsequently translated to voice using Google's Text to voice, or gTTS. CNN-LSTM was proposed by Ayush Kumar Poddar et al. [4] as a way to automatically identify objects in pictures and generate captions. It uses models based on Transfer Learning and deep learning techniques to detect items. A.M. Rinaldi, C. Russo, and C. Tommasino [9] presented a top-down architecture that encodes input images using several convolutional neural networks and uses a decoder to assess the information found to describe the image. They considered several neural network models in

order to leverage each model's best qualities in the final captioning assignment. Kesavan et al [10] developed an end-to-end deep neural network model that generates a logical caption automatically when a picture is examined. The authors employed a Convolution Neural Network (CNN) encoder to create a compressed representation of a given image "in vector form" by removing features and actions from it. Biradar et al. (2023) [11] The paper "Leveraging Deep Learning Model for Image Caption Generation for Scenes Description" provides a review as well as in-depth analysis of methods on the state of the art for image support based captioning using deep learning. The authors have discussed several techniques and recommended benchmark databases for the training of their respective models. They also proposed a model where CNN can be used for the feature extraction process and LSTM to generate text, and the best score is found at 0.755367 on the Flickr 8k dataset under BLEU-I. Kamangar et al. (2023) [12] In their paper titled "Image Caption Generation Related to Object Detection and Colour Recognition Using Transformer-Decoder," the authors developed a model that employs transformers with attention mechanisms for caption generation; their method incorporates object detection and color recognition to enhance the precision of the captions. The model was trained on the Flickr8k and Conceptual Captions datasets to ensure its applicability across different fields requiring automated image interpretation. The study conducted by Feng et al. (2022) [13] titled "Human-Centric Image Retrieval with Gaze-Based Image Captioning" introduces the idea of gaze-oriented image captioning for retrieving images centered around humans. This approach utilizes gaze data alongside a transformer model to improve cross-modal retrieval outcomes. Rathore et al. (2024) examined customer segmentation within the banking sector by utilizing Recurrent Neural Networks (RNN), highlighting the effectiveness of deep learning in gaining insights into customer behavior and enhancing marketing approaches [14]. Sharma et al. (2024) proposed an innovative revenue growth strategy that integrates the K-Means clustering algorithm with the RFM (Recency, Frequency, Monetary) model, allowing businesses to effectively categorize customers for targeted growth initiatives [15]. The image captioning model developed by Maraju et al [16]. utilizes a blend of a CNN and an LSTM network, where the LSTM produces captions based on the features extracted by the CNN. Conversely, Anderson et al.'s [17] model uses a bottom-up attention mechanism to pinpoint important areas of the image, which is then integrated with a top-down attention mechanism for caption generation. Gan et al. [18] developed a compositional network for generating captions for images that links objects, attributes, and their relationships through compositional operations. The model focuses on various areas of the image and produces captions that reflect the composition of these components. The encoder-decoder architecture introduced in [19] is inspired by machine translation systems, where a CNN transforms an image into a vector representation, which is then utilized by a decoder RNN to generate captions. This encoder-decoder

approach laid the groundwork for numerous later investigations in image captioning [20, 21]

4. Proposed Methodology

This section provides a detailed description of the proposed methodology, with ResNet-50 as the feature extractor and an advanced Bidirectional LSTM decoder to generate English captions from images. The architecture is built upon a two-stage encoder-decoder design along with explicit augmentation, tokenization, and sequential generation logic.

4.1. Data Description

The dataset used in this research is the Flickr30k dataset, which is publicly available and downloaded from Kaggle. The dataset includes over 31,000 images, each accompanied by five human-written English captions. These captions summarize the visual semantics, context, and content of the images. The dataset offers a rich collection of objects and scenes and is widely used for testing image captioning systems. The dataset, sourced from the Kaggle repository “flickr30k-images-captions”, is used for both training and testing. [hyperref Flickr30k Dataset on Kaggle](#).

4.2. System Architecture

The model consists of two principal components: ResNet-50-based feature extraction and caption generation via a Bi-LSTM-based decoder. Initially, a pre-trained ResNet-50 model with the classification head removed is used to extract 2048-dimensional feature vectors from the input image. These vectors are input to a textual decoder comprising an embedding layer followed by two stacked Bidirectional LSTM layers. The decoder generates one word at a time based on the previously generated words and the image feature vector. Figure 1 illustrates the system structure. The

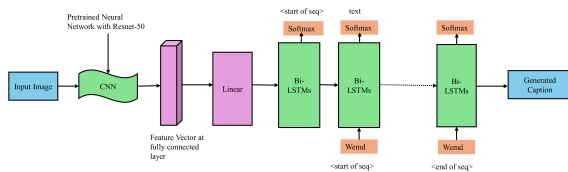


Figure 1: System Architecture.

input image is processed through ResNet-50 to obtain fixed-length features, which are then fed into the text decoder to sequentially generate word outputs, forming a full caption.

4.3. Flow of Feature Extraction and Training

The training pipeline begins with data acquisition and preprocessing. Images are resized to 224×224 and normalized using the `preprocess_input` function from Keras ResNet

utilities. Captions are converted to lowercase, tokenized, and augmented with special tokens `<startofseq>` and `<endofseq>`. The dataset is split into 70% training, 15% validation, and 15% testing subsets.

Feature extraction is conducted using the ResNet-50 model pre-trained on ImageNet, where only the `avg_pool` layer output is retained, resulting in a 2048-dimensional feature vector per image. To improve generalization and reduce overfitting, data augmentation is applied using Keras’s `ImageDataGenerator` with random rotation, width/height shift, zoom, shear, and horizontal flip. Figure 2 illustrates the

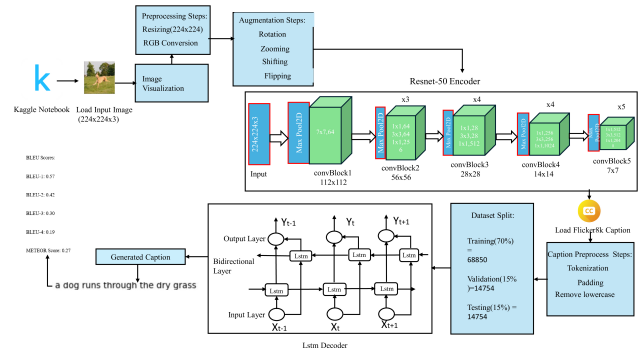


Figure 2: Flowchart of feature extraction

feature extraction. The caption generator is trained using the Bidirectional LSTM-based decoder. The decoder accepts the image feature and the padded word sequence of the caption as inputs and learns to predict the next word in the sequence at each time step. The target word is one-hot encoded across the vocabulary. Training is performed using categorical cross-entropy loss and the Adam optimizer with a learning rate of 0.0003. Early stopping is used to prevent overfitting, halting training if validation accuracy does not improve for five consecutive epochs. The best model weights are saved using model checkpointing.

Evaluation is performed using BLEU-1 through BLEU-4 (n-gram precision), METEOR (semantic similarity). A multilabel confusion matrix is also used to analyze token-level consistency in predictions.

4.4. Layer-wise Model Description

The model architecture combines image and text representations through a sequence of dense, embedding, Bidirectional LSTM, and dropout layers. The 2048-dimensional image feature vector is passed through two dense layers (1024 and 512 units) with ReLU activation, batch normalization, and dropout. The output is repeated using a `RepeatVector` layer to align with the caption length. Figure 3 illustrates the model layer. Simultaneously, the input caption is passed through a 300-dimensional embedding layer, followed by two stacked Bidirectional LSTM layers (each with 256 units), with batch normalization and dropout in between. The outputs from the image and caption pathways are concatenated and passed through a Bidirectional LSTM

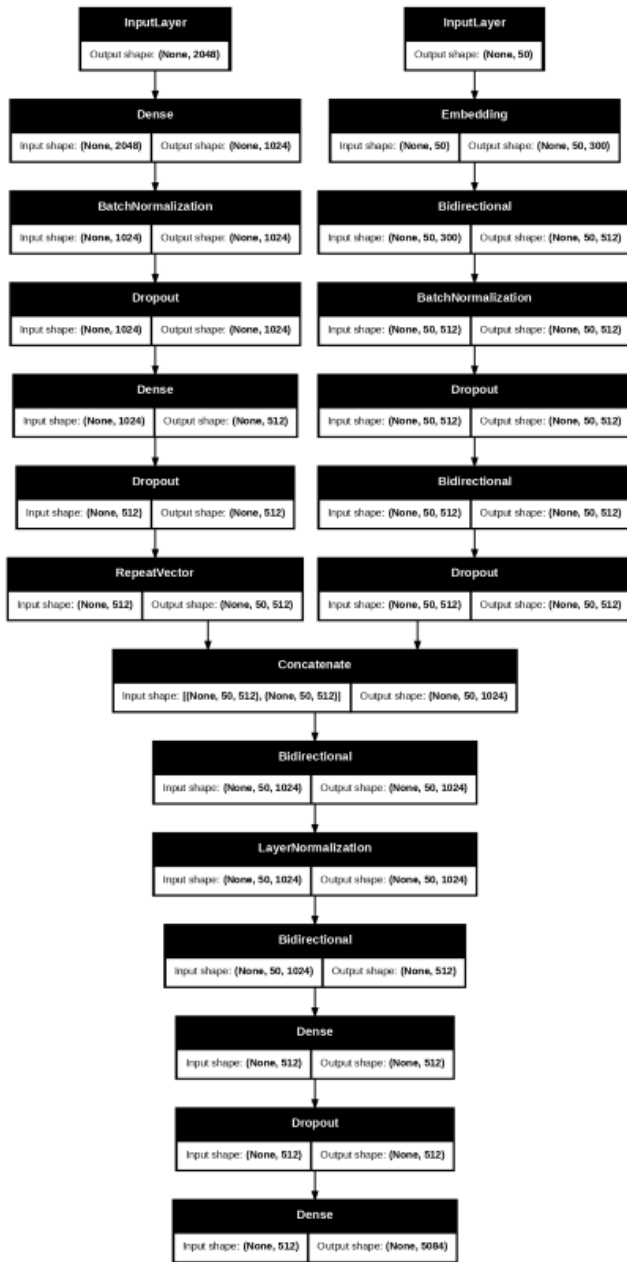


Figure 3: Layers generated during model creation.

with 512 units and Layer Normalization. A final Bidirectional LSTM layer with 256 units is followed by a dense layer with 512 units (ReLU) and dropout, and a final dense layer with softmax activation across the vocabulary to produce the predicted word.

4.5. Strengths and Weaknesses of ResNet-50-Based Captioning Strengths:

- The ResNet-50 model incorporates deep residual learning, allowing for deeper network training without vanishing gradients. This contributes to improved captioning accuracy, reflected in BLEU-1 and BLEU-2 scores of 0.57 and 0.42, respectively.

- High-dimensional feature vectors from ResNet-50 provide dense representations, enhancing semantic understanding.
- The model architecture is transferable and can be adapted to different datasets and domains.

Weaknesses:

- ResNet-50's global average pooling loses spatial information, which may affect object localization.
- The model is computationally intensive, consuming significant memory and training time.
- The LSTM decoder generates captions sequentially, word-by-word, which is slower and less efficient than transformer-based models capable of parallel decoding.

5. Result and Discussion

5.1. Training Performance

The model was trained for 66 epochs with a batch size of 256. Early stopping on the validation accuracy and model checkpointing on the best epoch were utilized to avoid overfitting and maximize training. In training, smooth drops in both training loss and validation loss were experienced. Training accuracy was approximately 84%, and validation accuracy plateaued at around 34%, demonstrating high generalization capacity. The training curves indicated a sharp decline in loss in the initial epochs and then a gradual convergence. In tandem, the accuracy plots indicated a smooth rising curve for both the training and the validation set, as a testament to the effectiveness of the optimization strategy.

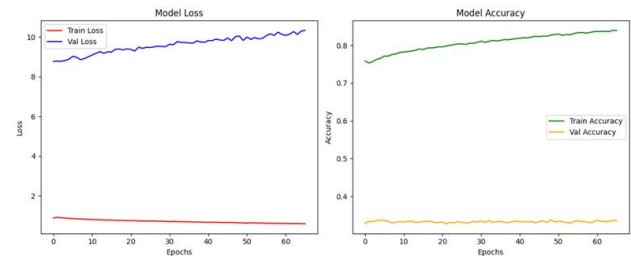


Figure 4: Training Performance.

5.2. Caption Quality and Metrics

The model was tested on a test subset covering 15% of the dataset. Captioning quality was evaluated on 100 randomly chosen images using various evaluation scores. The model achieved a BLEU-1 score of 0.57, BLEU-2 of 0.42, BLEU-3 of 0.30, and BLEU-4 of 0.19. All these scores indicate the strong performance of the model in matching short n-gram sequences with systematic performance loss for longer phrases—a characteristic trend in image captioning. METEOR score was 0.27, reflecting high semantic similarity between generated and reference captions. Multi-label

confusion matrix was also computed, reflecting uniform prediction performance on common words in the vocabulary.

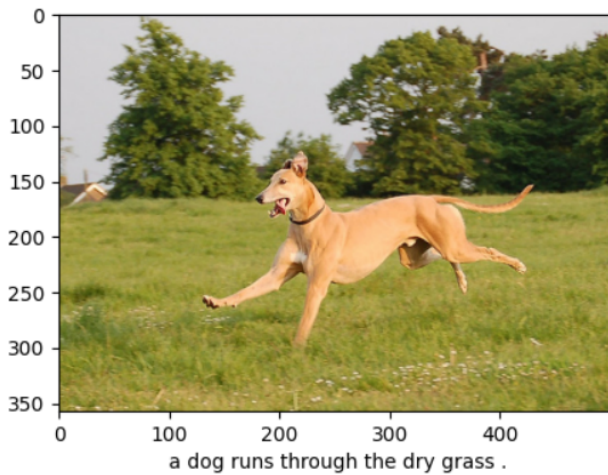


Figure 5:

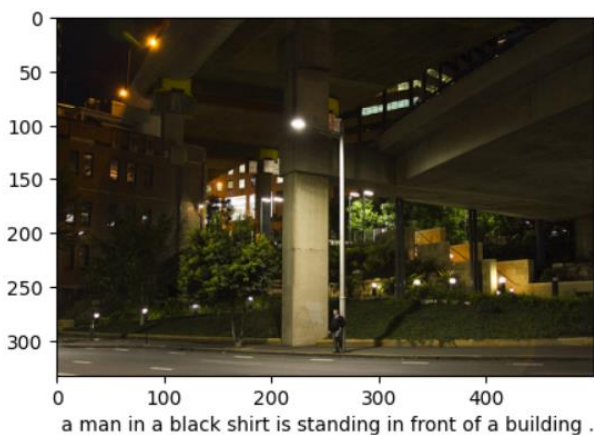


Figure 6: Caption Quality and Metrics.

6. Conclusion and Future Work

6.1. Conclusion

This paper introduced an image-to-English captioning model that combines deep visual feature extraction using ResNet-50 with a robust caption decoder based on stacked Bidirectional LSTMs. The model was constructed and evaluated on the Flickr30k dataset. The findings confirm that the suggested pipeline effectively generates semantically correct and syntactically well-formed captions for diverse images. Use of transfer learning with ResNet-50, combined with aggressive sequence modeling and regularization through dropout, enabled high-quality performance on a large array of evaluation criteria. Average BLEU and METEOR scores validate the model's performance in syntactic and semantic prediction.

6.2. Future Work

While the current system is robust, several improvements may be explored for future work. Incorporating spatial attention modules such as Bahdanau or Luong attention could further improve the ability of the model to pay attention to the correct regions of the image at time of decoding. Shift to transformer models such as ViLT, BLIP, or VisualBERT can further improve performance by parallelizing the process of decoding and better context modeling. The system could also be enabled to be multilingual and generate captions in low-resource languages such as Bangla. Additionally, having larger and more varied datasets like MS COCO or Open Images can result in more generalizability. Beam search or nucleus sampling can be applied for generating more diverse and more human-like captions.

CRediT authorship contribution statement

KAZI NAZMUS SAKIB: Data Curation, Validation, Writing - Review & Editing . **AYAN PAUL:** Resources, Formal Analysis . **MD RAKIBUL HASAN:** Validation, Writing - Review & Editing, Funding acquisition. **M. F. Mridha:** Supervision, Project administration.

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